Financial Analysis and Decision Making
Global BBA Fall 2026

Jason Kang

Session 8

BUSINESS SCHOOL

SKEMA

Housekeeping

About me

- Office: B3.09
- Office Hours: E-mail for appointment or walk-in for short queries
- jason.kang@skema.edu

Portfolio-in-Practice



In-class exam, November 19

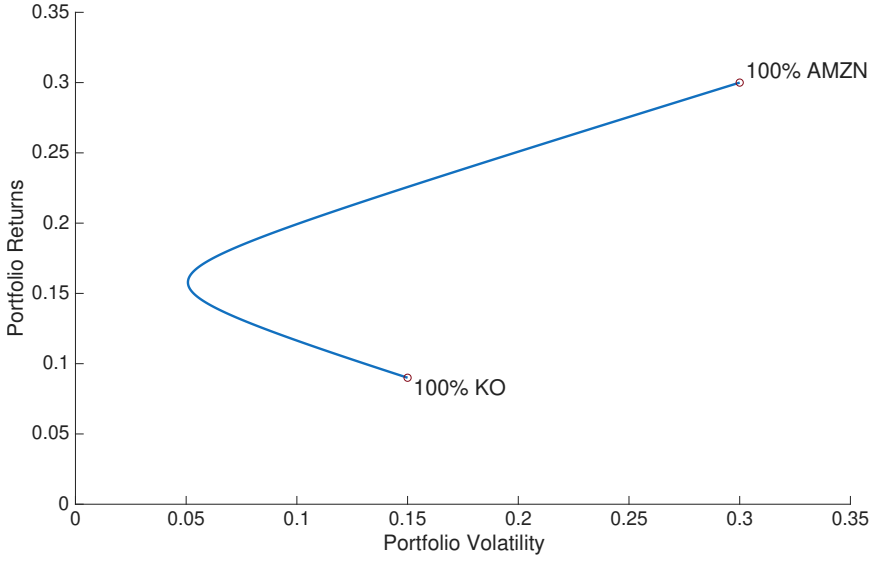
- FADM Chapters 6 & 7 (last week and this week)
- Roughly 60 minutes
- Random seating assignments

Plan For Today



- Review: Portfolio Diversification
- Capital Asset Pricing Model
- Computing Stock Returns

Review: The Efficient Frontier, $\rho = -0.87$



Portfolio Variance

Last week, we computed the variance of a portfolio in two ways:

$$\text{var}(p) = \sigma_p^2 = w^2\sigma_{\text{KO}}^2 + (1-w)^2\sigma_{\text{AMZN}}^2 + 2w(1-w)\text{cov}(\text{KO}, \text{AMZN})$$

and

$$\text{var}(p) = \sigma_p^2 = w^2\sigma_{\text{KO}}^2 + (1-w)^2\sigma_{\text{AMZN}}^2 + 2w(1-w)\rho\sigma_{\text{KO}}\sigma_{\text{AMZN}}$$

where w was the weight invested in KO.

Portfolio Variance

As the appendix slides show, our portfolio variance can be decomposed into variance and covariance components when we assume equal weights across all stocks in our portfolio.

$$\sigma_p^2 = \left(\frac{1}{N}\right)\bar{\sigma}^2 + \left(\frac{N-1}{N}\right)\overline{\text{cov}}$$

where

- $\bar{\sigma}^2$ is the average variance across all assets
- $\overline{\text{cov}}$ is the average covariance between all assets

*Try plugging in $w = 0.5$ in the previous slide. Both approaches gives you $\sigma_p = 0.09287$.

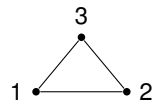
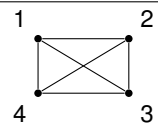
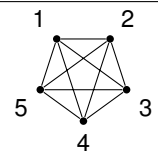
Portfolio Diversification

Suppose we consider large N. Taking limits as $N \to \infty$

$$\begin{aligned}\lim_{N\to\infty} \sigma_p^2 &= \lim_{N\to\infty} \left[\left(\frac{1}{N} \right) \bar{\sigma}^2 + \left(\frac{N-1}{N} \right) \overline{\text{cov}} \right] \\ &= \overline{\text{cov}}\end{aligned}$$

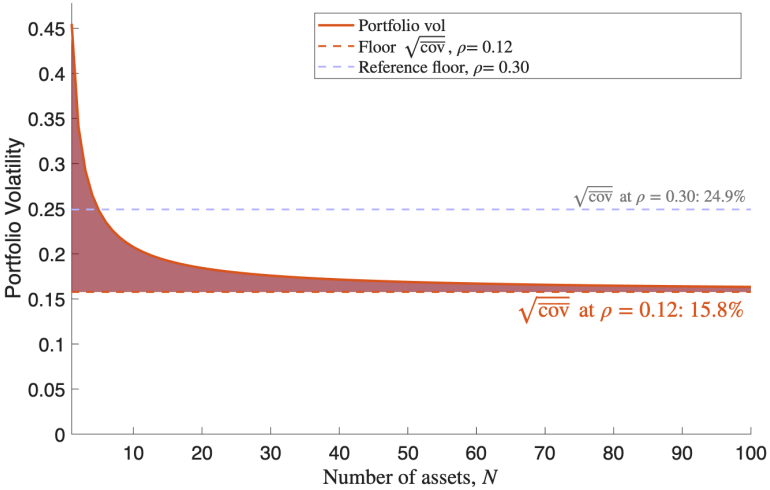
Our portfolio variance is determined by the average covariance between all assets as N becomes large.

Recall: Portfolio Variance with Many Assets

2 Assets		1 distinct **correlation**
3 Assets		3 distinct **correlations**
4 Assets		6 distinct **correlations**
5 Assets		10 distinct **correlations**

For N assets, there are $N(N-1)/2$ correlation pairs. When $N = 100$, there are 100 variance terms, but 4950 covariance terms!

Portfolio Volatility: $\bar{\sigma} = 0.45,\ \bar{\rho} = 0.12$



Market Risk vs Idiosyncratic Risk



There are many names, but firm-specific, idiosyncratic, or diversifiable risk refers to good or bad news about the company itself.

Example. The risk of a CEO targeted by a foreign intelligence officer in an extramarital affair.

An investor, by holding a diversified portfolio of many stocks, can reduce his/her portfolio volatility to $\sqrt{\overline{\text{cov}}}$. We call the risk that investors cannot remove as market, systematic, or non-diversifiable risk.

Namely, these are events/news that contribute to the covariance between two assets.

Which type of risk is this?

- Covid-19
- A global recession / financial crises
- Failed clinical trial
- Product failure or recall
- Lawsuits
- Interest rate hikes
- Earthquakes
- New product launch
- CEO scandal

Plan For Today



- Review: Portfolio Diversification
- Capital Asset Pricing Model
- Computing Stock Returns

Compensation for Taking Risk



If an investor has a diversified portfolio of many stocks, adding one more stock only contributes to the average covariance of the portfolio, as its individual volatility largely washes out (diversifiable risk).

If the stock does well when the rest of your portfolio does well, this stock amplifies the risk of your portfolio.

If the stock does well when the rest of your portfolio does poorly, this stock reduces the risk of your portfolio by acting as insurance.

Compensation for Taking Risk



If the stock's own volatility does not contribute much towards the risk of the diversified portfolio, this means that we do not care about the total risk (volatility) of an individual stock, but only its covariance to our portfolio.

Recall, the two-fund separation theorem says investors should hold the market portfolio as it gives the best ratio of excess returns per unit of risk.
$\Leftrightarrow$ We should care about the stock's covariance to the market portfolio.

Measure of Covariance

The beta, or β, of an asset measures its sensitivity to market risk.

For stock *i*, its beta is

$$\beta^i = \frac{\text{cov}(R_i, R_m)}{\text{var}(R_m)} = \rho_{i,m} \times \frac{\sigma_i}{\sigma_m}$$

where

- R_i is the return on asset *i*
- R_m is the return on the market portfolio
- $\rho_{i,m}$ is the correlation between asset *i* and the market
- $\sigma_i,\ \sigma_m$ are the volatilities of asset *i* and the market

Measure of Covariance

The beta, or β, of an asset measures its sensitivity to market risk.

For stock *i*, its beta is

$$\beta^i = \frac{\text{cov}(R_i, R_m)}{\text{var}(R_m)} = \rho_{i,m} \times \frac{\sigma_i}{\sigma_m}$$

Example. Suppose the correlation between KO and the market is equal to 0.33. KO has an annual volatility of 15%, and the market portfolio has an annual volatility of 18%. What is the beta of KO?

Measure of Covariance



Interpretation.

If the market portfolio goes up by 1 percentage point, we expect the stock to go up by β percentage points.

Example. Suppose the S&P 500 went up by 3%. By what percent do we expect the stock price of KO to go up by?

Beta's of Individual Companies

Based on 5-year daily data, 2021-2025.

Ticker	β	Ticker	β
JNJ	0.18	MSFT	1.09
CL	0.19	AAPL	1.17
PG	0.27	NFLX	1.26
KO	0.28	AMZN	1.43
MCD	0.38	UBER	1.44
SBUX	0.96	TSLA	2.03
DIS	1.00	NVDA	2.08
NKE	1.06		

The Capital Asset Pricing Model (CAPM)



The CAPM states that an individual company i's expected return can be completely explained by its covariance to the market portfolio.

$$\mathbb{E}[R_i] = R_f + \beta_i \times \underbrace{(\mathbb{E}[R_m] - R_f)}_{\text{risk premium}}$$

We call $\mathbb{E}[R_m] - R_f$ the market risk premium. It is the compensation for bearing risk, the expected return in excess of holding the risk-free asset.

Example. What is $\mathbb{E}[R_i]$ if

- $\beta_i = 0$
- $\beta_i = 1$

Interpreting CAPM



An alternative interpretation of the CAPM is the required rate of return, or a company's equity cost of capital.

* As we have stressed repeatedly, investors only care about covariance to non-diversifiable risk, and not the total risk of individual stocks. The CAPM states that companies with identical market risk share the same equity cost of capital (what we use to discount cash flows by).

Formulas (so far)

The CAPM equation

$$\mathbb{E}[R_i] = R_f + \beta_i \times (\mathbb{E}[R_m] - R_f)$$

For stock i, its beta is

$$\beta^i = \frac{\text{cov}(R_i, R_m)}{\text{var}(R_m)} = \rho_{i,m} \times \frac{\sigma_i}{\sigma_m}$$

Example

Suppose the risk-free rate is 4.5%, and the expected return on the market is 12.5%.

a. Suppose the beta of MCD is 0.38. What is the expected return on MCD stock?
   7.54%

Example

Suppose the risk-free rate is 4.5%, and the expected return on the market is 12.5%.

b. Suppose the correlation between NKE and the market portfolio 0.54 and the volatility of NKE is 34.9%. If the volatility on the market portfolio is 17.7%, compute
    i The beta of NKE
    ii The expected return on NKE

Example

Suppose the risk-free rate is 4.5%, and the expected return on the market is 12.5%.

c. Assume the volatility on the market portfolio is 17.7% as before. If the beta of NFLX is 1.26, and the correlation between NFLX and the market portfolio is 0.51, compute the annual volatility of NFLX stock.

Example

Suppose the risk-free rate is 4.5%, and the market risk premium is 5%.

d. What is the required rate of return on investing in Disney. which has a $\beta_{\text{DIS}} = 1.0$?

Example



Suppose the risk-free rate is 4.5%, and the expected return on the market is 12.5%.

e. What is the required rate of return on investing in gold. which has a $\beta = -0.5$?

The Security Market Line (SML)

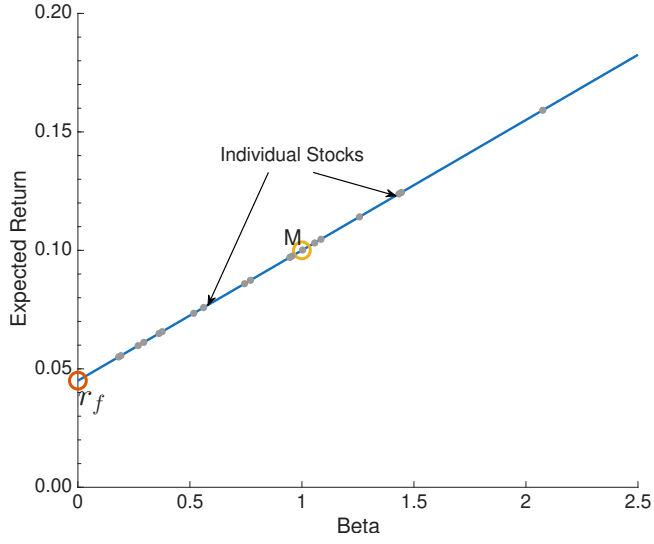


The CAPM is a strong statement. It says that the expected returns of an individual stock is completely captured by its market beta.
$\rightarrow$If AMZN has higher stock returns than KO, then it is because AMZN has higher market beta.

Graphically, the security market line (SML) plots the expected returns of individual stocks against their beta's.

According to the CAPM, the expected returns of all stocks should lie along a single line.

The Security Market Line



The Security Market Line

The formula for the SML is simply the CAPM,

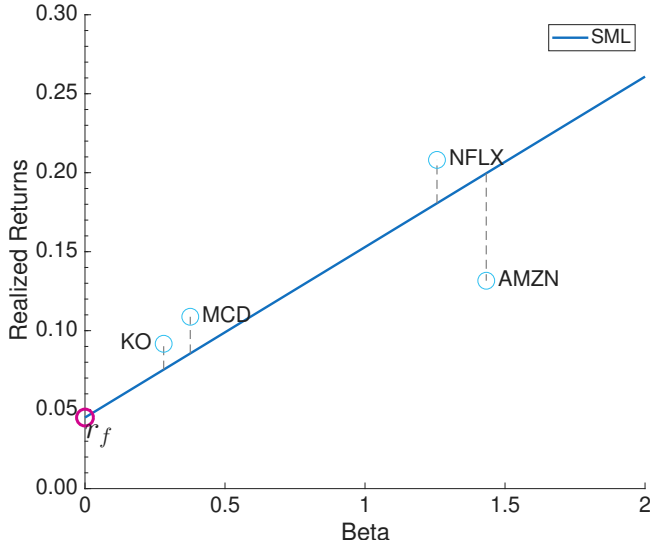
$$\mathbb{E}[R_i] = R_f + \beta_i \times (\mathbb{E}[R_m] - R_f)$$

What is the slope of the security market line?

What is the y-intercept?

How can you infer the market risk premium based on the graph of the SML?

The Security Market Line: Realized vs Expected



Measuring Performance

The CAPM is a model of expected returns but the previous plot showed that realized returns do not lie exactly on the SML. Does this mean the CAPM is wrong?

Define alpha as a measure of performance in excess of what is predicted/required by the CAPM,

$$\alpha_i \doteq R_i - \underbrace{\left[R_f + \beta_i \times (\mathbb{E}[R_m] - R_f)\right]}_{\text{CAPM}}$$

$\alpha > 0$ implies ?

Measuring Performance

Most hedge funds and mutual fund managers talk about positive alpha as a sign of "beating the market" because you are delivering returns higher than the fair compensation required for the level of risk you are taking.

Example. Suppose our portfolio has a beta of 1.5. The risk-free rate is 4.5% and the market risk premium is 5.5%. If our portfolio returns were 14% over the past year, what was the alpha of our portfolio, and how was our performance relative to what is predicted by the CAPM?

Measuring Performance



Again, the CAPM is a model of expected returns, so some stocks and portfolios earning positive/negative alpha is not a violation of CAPM. Remember, individual stocks carry idiosyncratic volatility (firm-specific news) that largely gets washed away in a diversified portfolio.

It becomes a problem if alpha's across all assets are not jointly zero, as it means the CAPM is systematically failing to capture an aspect of expected returns.

Example

- Suppose low beta stocks persistently have positive alpha year after year.

Portfolio Beta

The beta of a portfolio is just the beta of the individual assets weighted by their portfolio weights,

$$\beta_p = \sum_i w_i \beta_i$$

a. Suppose I have a portfolio of 25% in NVDA, 20% in AAPL, and 55% in MCD. The beta of each stock is $\beta_{\text{NVDA}} = 2.08, \beta_{\text{AAPL}} = 1.17, \beta_{\text{MCD}} = 0.38$ respectively. What is the beta of the portfolio?

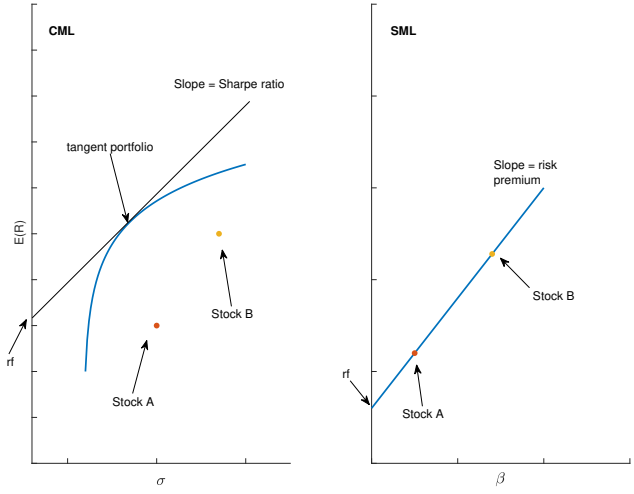
Portfolio Beta

The beta of a portfolio is just the beta of the individual assets weighted by their portfolio weights,

$$\beta_p = \sum_i w_i \beta_i$$

b. Suppose I have a portfolio of 60% in the market index (S&P 500), and 40% in the risk-free asset (a 10-year government bond). What is the beta of this portfolio?

CML versus SML



Wrapping Up



Don’t confuse the CML with the SML (think about what’s on the x-axis).

Importantly, the CAPM is not “the truth”, but it is a useful framework to think about risk and return.

Even today, virtually all valuations rely on the CAPM to measure a company’s cost of equity.

Exercise 3 (FADM Ch.7 Slide 35)



Suppose the stock of the 3M Company (MMM) has a beta of 0.69 and the beta of Hewlett-Packard Co. (HPQ) stock is 1.77. Assume the risk-free interest rate is 5% and the expected return of the market portfolio is 12%. What is the expected return of a portfolio of 40% of 3M stock and 60% Hewlett-Packard stock, according to the CAPM?

14.37%

Exercise 11.49 (FADM Ch.7 Slide 36)

	Portfolio Weight	Volatility	Correlation with the Market Portfolio
HEC Corp	0.26	13%	0.35
Green Midget	0.29	28%	0.52
Alive And Well	0.45	11%	0.54

The volatility of the market portfolio is 10% and it has an expected return of 8%. The risk-free rate is 3%.

a. Compute the beta and expected return of each stock.
b. Using your answer from part a, calculate the expected return of the portfolio.
c. What is the beta of the portfolio?
d. Using your answer from part c, calculate the expected return of the portfolio and verify that it matches your answer to part b.

Exercise 11.49 (FADM Ch.7 Slide 36)

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The volatility of the market portfolio is 10% and it has an expected return of 8%. The risk-free rate is 3%.

a. Compute the beta and expected return of each stock.
   $\beta_{HEC} = 0.455, \beta_{GM} = 1.456, \beta_A = 0.594$
   $\mathbb{E}[R_{HEC}] = 5.28\%, \mathbb{E}[R_GM] = 10.28\%, \mathbb{E}[R_A] = 5.97\%.$
b. Using your answer from part a, calculate the expected return of the portfolio.
   $\mathbb{E}[R_p] = 7.04\%.$
c. What is the beta of the portfolio? $\beta_p = 0.8078.$
d. Using your answer from part c, calculate the expected return of the portfolio and verify that it matches your answer to part b. $\mathbb{E}[R_p] = 7.04\%.$

Plan For Today



- Review: Portfolio Diversification
- Capital Asset Pricing Model
- Computing Stock Returns

Computing Stock Returns



In Session 6, we discussed equity financing as an alternative to debt. In addition to voting rights and the ability to trade a company's stock in the stock exchange, companies often pay dividends to shareholders.

Roughly 25% of U.S. firms pay regular dividends (typically paid quarterly).

- Mature, stable firms often return cash to investors using dividends
- Dividends signal confidence in the firm's ability to generate cash regularly
- It also signals that the firm does not have better uses of its cash

Who Pays Dividends?

As of FY 2025

Sector	*N*	% Pay Dividend	% Buyback	% Either
Consumer Discretionary	594	39.4	65.8	72.7
Consumer Staples	142	52.1	56.3	71.8
Communication Services	128	42.2	57.0	65.6
Industrials	929	39.4	60.5	67.5
Technology	1015	17.3	56.8	60.3
Energy & Materials	974	29.0	36.2	43.5
Healthcare	1070	6.7	33.8	35.0
Total (pooled)	**4,852**	**25.9**	**49.4**	**54.7**

Putting Names to Faces

Sector	Top players
Technology	SAP, Nvidia, Apple, Microsoft, IBM
Healthcare	Novo Nordisk, Roche, AstraZeneca, Novartis, Sanofi
Energy & Materials	Shell, TotalEnergies, L'Oréal, Unilever, L'Air Liquide
Industrials	Air France-KLM, Caterpillar, Boeing, Accenture
Consumer Discretionary	LVMH, Toyota, Amazon, Tesla, McDonald's
Consumer Staples	Nestlé, Danone, Heineken, Philip Morris, Coca-Cola
Communication Services	Deutsche Telekom, Vodafone, Orange, Netflix, Disney

Computing Stock Returns

Given the price P_{t+1} and dividends paid D_{t+1} at $t+1$, the stock return from t to $t+1$ is computed as

$$
\begin{aligned}
r_{t+1} &= \frac{D_{t+1} + P_{t+1} - P_t}{P_t} \\
&= \underbrace{\frac{D_{t+1}}{P_t}}_{\text{div. yield}} + \underbrace{\frac{P_{t+1} - P_t}{P_t}}_{\text{capital gains}}
\end{aligned}
$$

This is the net return, a number like 5%, in contrast to the *gross* return, $R_{t+1} = 1 + r_{t+1} = 1.05$

Computing Stock Returns

Date	Closing price	Dividend
11/11/2026	335.20	–
12/11/2026	338.30	2.00

Compute the daily stock returns on 12/11/2026 using

$$r_{t+1} = \underbrace{\frac{D_{t+1}}{P_t}}_{\text{div. yield}} + \underbrace{\frac{P_{t+1} - P_t}{P_t}}_{\text{capital gains}}$$

Computing Stock Returns

Suppose we have the following table of daily stock prices and returns

Date	Closing price	Dividend	Returns
08/11/2026	331	–	
09/11/2026	335	–	0.01208
10/11/2026	334	–	-0.00299

Compute the two-day stock returns from 08/11/2026 to 10/11/2026.

.00906

Computing Stock Returns



Things are more complicated when we include dividends.

Date	Closing price	Dividend	Returns
31/07/2026	100	–	
31/08/2026	105	–	0.05000
30/09/2026	112	2.00	0.08571
31/10/2026	85	–	-0.24107

Compute the quarterly stock returns from 31/07/2026 to 31/10/2026.

Computing Stock Returns (Technicalities)



Dividends can be re-invested into the stock at the time of payment.

Then, gross returns from date 0 to k are

$$R_{0\to k} \doteq 1 + r_{0\to k} = (1 + r_{0\to 1})(1 + r_{1\to 2}) \cdots (1 + r_{k-2\to k-1})(1 + r_{k-1\to k})$$

Computing Stock Returns (Technicalities)



Assume there are no dividends, or that prices are dividend-adjusted. Then, you can just divide the prices. For example, the gross return from date 0 to k is

$$R_{0\to k} = \prod_{j=0}^{k-1}(1 + r_{j\to j+1}) = \frac{P_1}{P_0}\frac{P_2}{P_1}\frac{P_3}{P_2}\frac{P_4}{P_3}\cdots\frac{P_k}{P_{k-1}} = \frac{P_k}{P_0}$$

The annual gross return would be dividing Dec 31_{2026} by Dec 31_{2025} prices (for example).

Just FYI, most data vendors give you the "adjusted close" price which allow you to just divide prices without worrying about dividends.



Appendix (optional material, can skip)

Derivation: Limit Results on Portfolio Diversification



Recall,

$$\sigma_p^2 = \text{var}(aX + bY) = a^2\text{var}(X) + b^2\text{var}(Y) + 2ab\text{cov}(X, Y)$$

More generally for N assets,

$$\sigma_p^2 = \sum_i \sum_j w_i w_j \sigma_{ij}$$

for $i = 1...N$ and $j = 1..N$, where σ_{ij} is the covariance between asset i and asset j.

When $i = j$, it is just the variance of the asset.

Derivation: Limit Results on Portfolio Diversification



Assume an equal-weighted portfolio, $w_i = w_j = 1/N$,

$$\sigma_p^2 = \frac{1}{N^2} \sum_i \sum_j \sigma_{ij}$$

for $i = 1...N$ and $j = 1..N$.

There are N times that $i = j$, so there are N variance terms.
There are $N(N-1)$ times that $i \neq j$, so there are $N(N-1)$ covariance terms.

Derivation: Limit Results on Portfolio Diversification



Splitting into a variance component and a covariance component,

$$\sigma_p^2 = \frac{1}{N^2}\left(\sum_{i=j}\sigma_{ij} + \sum_{i\neq j}\sigma_{ij}\right)$$

Defining $\bar{\sigma}^2 \doteq \frac{1}{N}\sum_{i=j}\sigma_{ij}$ as the average variance,
and $\overline{\text{cov}} \doteq \frac{1}{N(N-1)}\sum_{i\neq j}\sigma_{ij}$ as the average covariance,

$$\begin{aligned}\sigma_p^2 &= \frac{1}{N^2}\left(N\bar{\sigma}^2 + N(N-1)\overline{\text{cov}}\right)\\ &= \left(\frac{1}{N}\right)\bar{\sigma}^2 + \left(\frac{N-1}{N}\right)\overline{\text{cov}}\end{aligned}$$

Derivation: Limit Results on Portfolio Diversification

Our portfolio variance is

$$\sigma_p^2 = \left(\frac{1}{N}\right)\bar{\sigma}^2 + \left(\frac{N-1}{N}\right)\overline{\text{cov}}$$

Taking limits as $N \to \infty$

$$\lim_{N\to\infty} \sigma_p^2 = \overline{\text{cov}}$$

Our portfolio variance is determined by the average covariance between all assets as N becomes large.

CAPM in practice

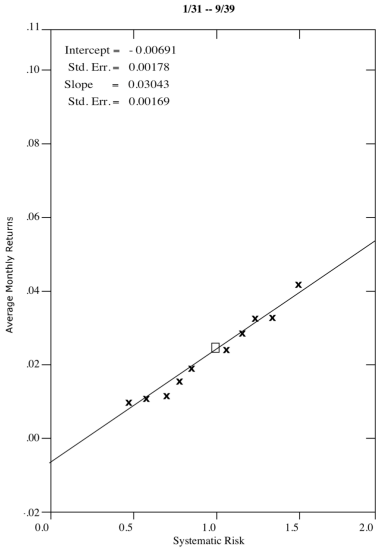


Figure 8 Average Excess monthly returns versus systematic risk for the 105 -month period January, 1931-September, 1939.

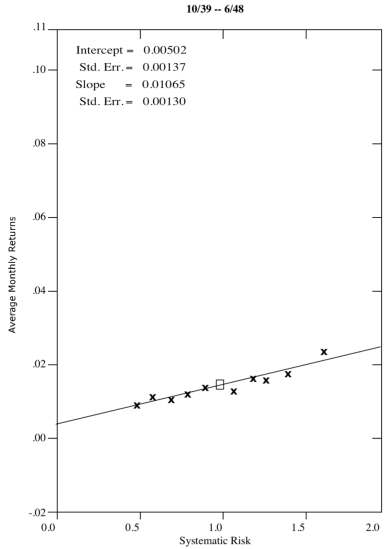


Figure 9 Average Excess monthly returns versus systematic risk for the 105 - month period October, 1939-June, 1948.

CAPM in practice is too flat

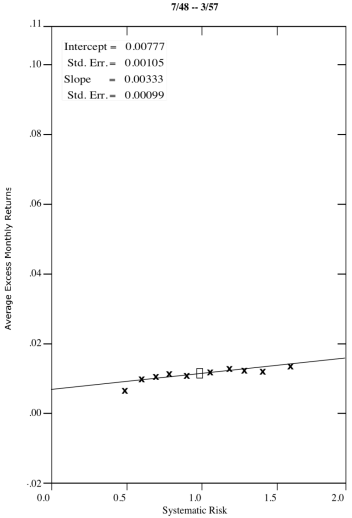


Figure 4 Average Excess monthly returns versus systematic risk for the 105 -month period July, 1948-March, 1957. Symbols as in Figure 1.

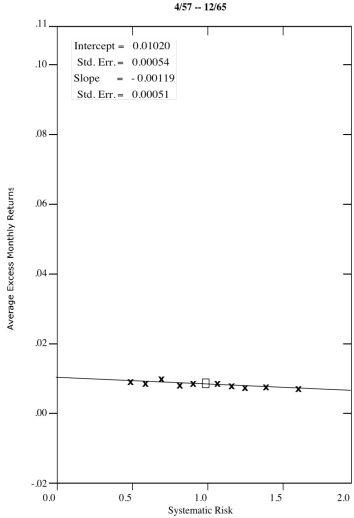
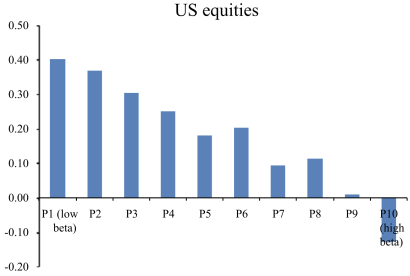
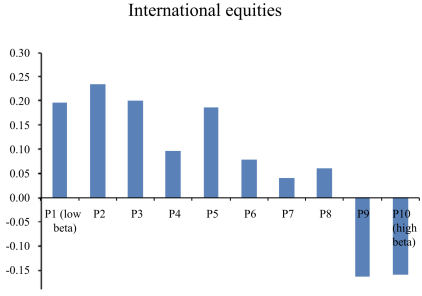


Figure 4 Average Excess monthly returns versus systematic risk for the 105 -month period April, 1957 - December, 1965. Symbols as in Figure 1.

CAPM in practice is too flat, even in 2014

- Average monthly alpha, in % for 10 portfolios sorted on CAPM beta.

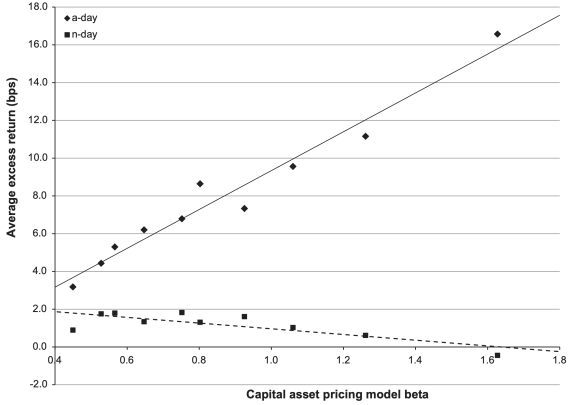




Source: Frazzini, Andrea and Pedersen, Lasse Heje, 2014, Betting against beta, *Journal of Financial Economics* 111, 1-25.

CAPM in practice is too flat, even in 2014, unless you look at the right dates.

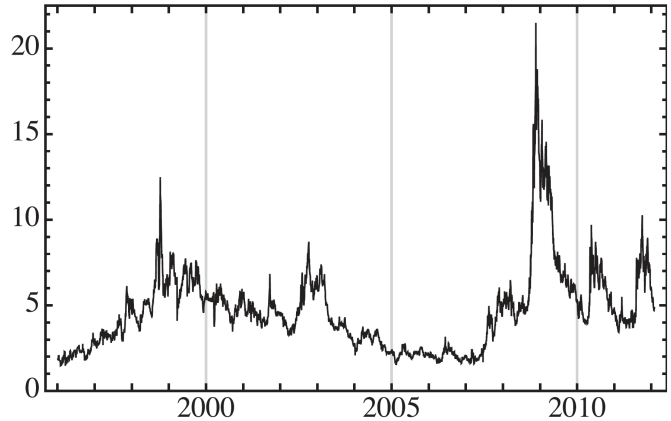
- CAPM fit on Federal Open Market Committee, inflation & unemployment data announcement days.



Source: Savor, Pavel and Wilson, Mungo, 2014, Asset pricing: A tale of two days, *Journal of Financial Economics* 113, 171-201.

But which market risk premium to use is also problematic.

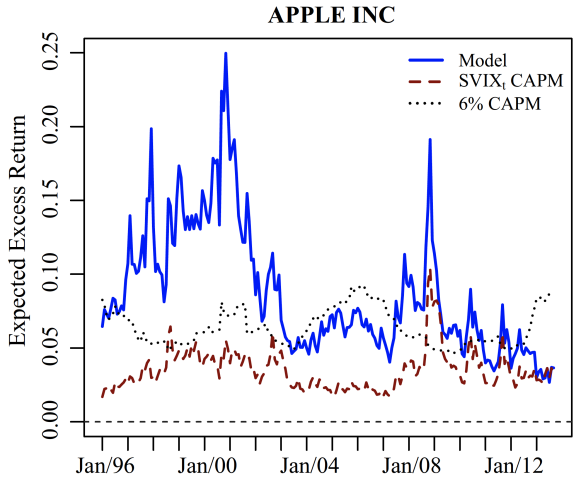
- Lower bound on the equity risk premium implied by option prices. It's not always 6%, sometimes it's 20%!



Source: Martin, Ian, 2017, What is the expected return on the market?, *The Quarterly Journal of Economics* 132(1), 367-433.

So what is a firm's cost of equity?

- Apple Inc.'s cost of equity, implied by option prices, in blue.



Source: Martin, Ian and Wagner, Christian, 2019, What is the expected return on a stock?, *The Journal of Finance* 74(4), 1887-1929.